

1.

设备名称	Lukas的宇宙飞船
处理器	13th Gen Intel(R) Core(TM) i7-13700H 2.40 GHz
机带 RAM	16.0 GB (15.6 GB 可用)
设备 ID	9DDF762D-9240-4580-B15B-E4CAFB8B998F
产品 ID	00342-30988-11979-AAOEM
系统类型	64 位操作系统, 基于 x64 的处理器
笔和触控	没有可用于此显示器的笔或触控输入

2.可运行 864 个 test

```
=====
||Ax-b||_oo/(eps*(||A||_oo*||x||_oo+||b||_oo)*N)= 3.31464624e-02 ..... PASSED
=====

Finished      864 tests with the following results:
               864 tests completed and passed residual checks,
                 0 tests completed and failed residual checks,
                 0 tests skipped because of illegal input values.

=====

End of Tests.
=====
```

3. 操作系统版本, 5. 15. 153;

Mpi 版本 4. 0

```
lukas@Lukas:~/hpl/bin/linux$ mpirun --version
HYDRA build details:
  Version: 4.0
  Release Date: Fri Jan 21 10:42:29 CST 2022
  CC: gcc -Wdate-time -D_FORTIFY_SOURCE=2 -g -O2 -ffile-prefix-map=/build/mpich-0xgrG5/mpich-4.0=. -flto=auto -ffat-lto-objects -flto=auto -ffat-lto-objects -fstack-protector-strong -Wformat -Werror=format-security -Wl,-Bsymbolic-functions -flto=auto -ffat-lto-objects -flto=auto -Wl,-z,relro
  Configure options: '-with-hwloc-prefix=/usr' '-with-device=ch4:ofi' 'FFLAGS=-O2 -ffile-prefix-map=/build/mpich-0xgrG5/mpich-4.0=. -flto=auto -ffat-lto-objects -flto=auto -ffat-lto-objects -fstack-protector-strong -fallow-invalid-boz -fallow-argument-mismatch' '--prefix=/usr' 'CFLAGS=-g -O2 -ffile-prefix-map=/build/mpich-0xgrG5/mpich-4.0=. -flto=auto -ffat-lto-objects -flto=auto -ffat-lto-objects -fstack-protector-strong -Wformat -Werror=format-security' 'LDFLAGS=-Wl,-Bsymbolic-functions -flto=auto -ffat-lto-objects -flto=auto -Wl,-z,relro' 'CPPFLAGS=-Wdate-time -D_FORTIFY_SOURCE=2'
  Process Manager: pmi
  Launchers available: ssh rsh fork slurm ll lsf sge manual persist
  Topology libraries available: hwloc
  Resource management kernels available: user slurm ll lsf sge pbs cobalt
  Demux engines available: poll select

Linux Lukas 5.15.153.1-microsoft-standard-WSL2 #1 SMP Fri Mar 29 23:14:13 UTC 2024 x86_64 x86_64 x86_64 GNU/Linux
```

4. 将参数调成

- HPLinpack benchmark input file
- Innovative Computing Laboratory, University of Tennessee
- HPL.out output file name (if any)
- 6 device out (6=stdout,7=stderr,file)
- 4 # of problems sizes (N)

29 30 34 35	Ns
3	# of NBs
2 3 4	NBs
0	PMAP process mapping (0=Row-,1=Column-major)
3	# of process grids (P x Q)
2 4 5	Ps
10 5 4	Qs
16.0	threshold
3	# of panel fact
0 1 2	PFACTs (0=left, 1=Crout, 2=Right)
2	# of recursive stopping criterium
2 4	NBMINS (>= 1)
1	# of panels in recursion
2	NDIVs
3	# of recursive panel fact.
0 1 2	RFACTs (0=left, 1=Crout, 2=Right)
4	# of broadcast
0 1 2 3	BCASTs (0=1rg,1=1rM,2=2rg,3=2rM,4=Lng,5=LnM)
1	# of lookahead depth
0	DEPTHS (>=0)
2	SWAP (0=bin-exch,1=long,2=mix)
64	swapping threshold
0	L1 in (0=transposed,1=no-transposed) form
0	U in (0=transposed,1=no-transposed) form
1	Equilibration (0=no,1=yes)
8	memory alignment in double (> 0)

其中 WR01L2R4 N: 35 NB:4 P:2 Q:10

0.00 得到最大 gflops 值: 3.2206e-01

我先将测试结果输入到 hpl_output.txt 文件里，然后用 awk
 '/Gflops/ {getline; getline; print \$NF}' hpl_output.txt | sort -g |
 tail -1 打印出最大 gflops 值，然后用 awk '/3.2206e-01/ {print}'
 hpl_output.txt | sort -g | tail -1 得到最大 gflops 值对应的参数